

Highlights

- Open-source Python framework for forest estate decision support.
- Model I scheduling with transparent inputs and outputs.
- libCBM linkage for carbon stock and flux accounting.
- Hybrid aspatial-to-raster workflows with reproducible geospatial I/O.
- Deterministic reproduction package and archived scaling benchmarks.

Software availability

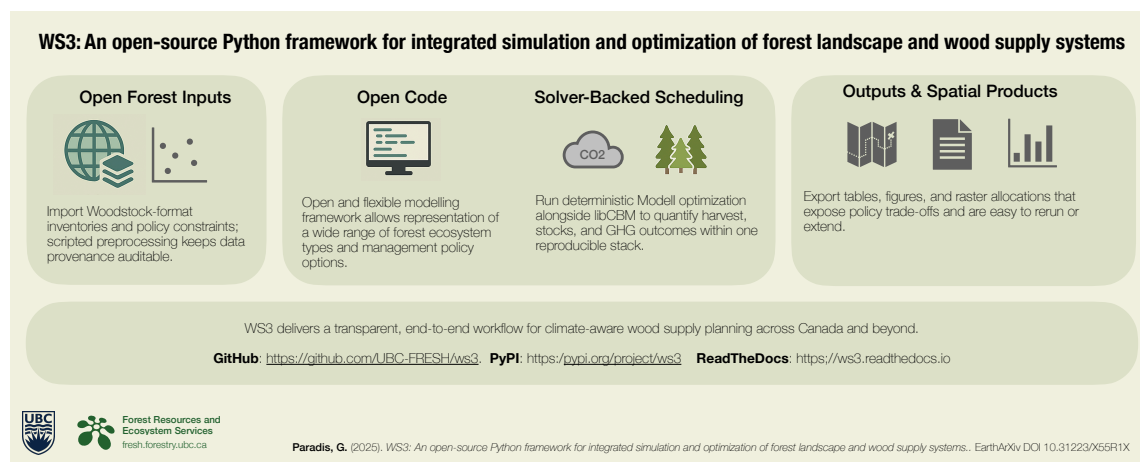
- Software name: WS3 (Wood Supply Simulation System)
- Developer and contact: Gregory Paradis (Faculty of Forestry, University of British Columbia); gregory.paradis@ubc.ca
- First public release and current version: 2015 (v0.1); current release v1.0.5
- Programming language and dependencies: Python (≥ 3.9); optional extras include `libcbm_py`, PuLP/HiGHS, and Gurobi
- Source repository and issue tracker: <https://github.com/UBC-FRESH/ws3>
- Distribution channel: PyPI package `ws3` (`pip install ws3` or `ws3[cbm]`) with wheels for Linux, macOS, and Windows
- Documentation: <https://ws3.readthedocs.io>
- Archived release: Zenodo DOI [10.5281/zenodo.17331213](https://doi.org/10.5281/zenodo.17331213)
- License: MIT
- Hardware and OS requirements: Linux, macOS, or Windows; tested on Ubuntu 24.04 and macOS 14; CPU-only with recommended 16+ GB RAM for large LP instances
- Tested Python versions: 3.9–3.12 (continuous-integration matrix)

- Data form and access: supplementary CSV, GeoTIFF, and notebook artifacts packaged in the GitHub repository and Zenodo release; total archive size < 50 MB excluding optional outputs

Graphical Abstract

WS3: An open-source Python framework for integrated simulation and optimization of forest landscape and wood supply systems

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WS3: An open-source Python framework for integrated simulation and optimization of forest landscape and wood supply systems

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Abstract

We describe WS3, an open-source Python framework that implements Model I scheduling, carbon accounting, and raster allocation in a single, scriptable workflow for reproducible forest planning analyses. WS3 ingests Woodstock-style inventories, actions, and scenarios; defines an explicit data model; and automates aspatial-spatial conversions so analysts can move from harvest schedules to georeferenced disturbance footprints. A documented linkage to the Canadian Forest Service Carbon Budget Model (libCBM) provides carbon stock and flux estimates for Canada and other libCBM-calibrated jurisdictions.

This paper documents a modular software architecture that decouples inventory ingestion, linear-programming harvest scheduling, carbon accounting, and raster allocation. Users can combine heuristic and solver-backed approaches and extend indicators. We provide a reproducible parity check against Woodstock on a toy dataset and report computational scaling experiments that characterize runtime and memory envelopes across desktop-to-cluster deployments. The complete toolchain—source code, configuration files, and deterministic reproduction scripts—is released under the MIT license with an archived Zenodo DOI.

We demonstrate WS3 on five timber supply areas in northeastern British Columbia, with individual inventories spanning 2.27 Mha (TSA 41) to 5.90 Mha (TSA 24) and a cumulative scaling run that aggregates all five to 20.40 Mha. Carbon trajectories are computed post hoc using libCBM to illustrate outcomes of alternative schedules, and raster allocation routines produce disturbance footprints that can be consumed by downstream spatial simulators such as SpaDES. The reproduction suite includes notebooks, container-ready scripts, and pinned dependency manifests so practitioners can adapt the workflow to their inventories and verify results against existing pipelines.

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WS3 provides an open implementation of workflows that are otherwise delivered through closed, licensed systems. The framework is intended as reusable informatics infrastructure for decision support under competing sustainability mandates.

Keywords: computational ecology, decision support, forest planning, optimization, carbon accounting, geospatial workflows

1. Introduction

Forested landscapes now sit at the centre of intertwined climate mitigation, biodiversity, and bioeconomy mandates, and planners are being asked to reconcile wood supply security with measurable reductions in greenhouse-gas emissions across multi-decadal horizons [3, 14, 10]. Meeting these expectations demands decision-support systems that can span harvest scheduling, regeneration and silviculture options, spatial footprint analysis, and explicit carbon accounting, while remaining transparent enough for regulators, Indigenous governments, and industrial partners to audit [25, 40]. The resulting analytical load has outgrown the ad hoc spreadsheets and siloed software stacks still common in practice, underscoring the need for integrated modelling platforms that support reproducible workflows and uphold open-science norms such as the PERFICT reproducibility principles for predictive ecology [19, 26, 27].

Strategic forest planning has long relied on operations-research formulations such as Model I path scheduling and its variants, which underpin allowable-cut analyses and forest estate studies worldwide [21, 15, 16, 35]. Model I is a strata-based linear program in which decision variables assign fractions of aggregated development-type strata to prescriptions over time; this enables large-scale policy analysis but necessarily abstracts away stand identities. In practice these ideas are delivered through proprietary toolchains—Woodstock, Patchworks, JLP (MELA), and Heureka among others—that package data models, solvers, and reporting inside closed desktop environments [38, 41, 29, 20]. While these systems are mature and widely trusted, closed licensing and opaque configuration formats can limit transparency, collaborative extension, and reproducible research workflows increasingly required by regulators and journals [19].

Open-source platforms have begun to address parts of this gap. Spatial simulation systems such as SpaDES and LANDIS-II, scenario-management frameworks like SyncroSim, and planning prototypes such as PRISM provide modular, shareable tooling [8, 24, 1, 31]. These systems excel at spatial dynamics and scenario orchestration but typically require users to supply their own planning modules, solver integrations, and carbon-accounting pipelines to assemble a full decision-support stack [9, 25]. Parallel efforts such as the French CAPSIS platform highlight the promise and complexity of interoperable forest growth modelling ecosystems [12, 11, 37]. This persistent gap between open landscape-simulation scaffolds and purpose-built forest-estate planning motivates the open, reproducible framework we introduce with WS3.

WS3 is a fully open Python framework that preserves the expressive power of legacy Model I

scheduling while embedding reproducibility, automation, and carbon accounting into the workflow. It imports Woodstock-format text files (or equivalent information in *ad hoc* formats via custom Python parsers) to protect historical investments, provides solver-flexible optimization, couples to the Canadian Forest Service libCBM implementation, and supports hybrid spatial–aspatial work-
flows compatible with geospatial rasters and SpaDES-based simulations [38, 28, 18, 17, 23, 4, 8]. We do not claim a new optimization formulation; rather, we provide a transparent and auditable implementation that can be inspected, extended, and reproduced across institutions [19, 26].

This article contributes to ecological informatics in three ways. First, it documents a modular decision-support architecture that combines harvest scheduling, carbon accounting, and spatial reporting in a single, scriptable workflow. Second, it provides evidence of functional parity with Woodstock on a toy dataset and reports computational scaling experiments that characterize run-time and memory envelopes for large landscapes. Third, it makes the complete toolchain—source code, configuration files, and reproduction scripts—available under an open license with an archived Zenodo DOI, enabling immediate reuse and extension by researchers and practitioners.

The remainder of this article details the WS3 architecture and data model (Section 2), demonstrates reproducible use-case workflows linking solver-backed scheduling to libCBM carbon accounting and spatial disturbance allocation (Section 3), discusses scope and limitations (Section 4), compares WS3 with alternative tools (Section 5), and summarises availability, authorship, and future work to facilitate uptake.

2. Software description

2.1. Architecture

WS3 is organized into five roles: (i) core data abstractions; (ii) forest model construction and simulation; (iii) optimization; (iv) spatial allocation; and (v) common utilities. The main modules are:

- forest: defines the ForestModel and core abstractions for development types (dtypes), actions, transitions, yields, and scenario compilation. A scenario is specified by a planning horizon, period length, an initial inventory (areas by dtype and age), and transition rules governing state changes under actions and growth.

- 60 • `opt`: provides a solver-flexible interface to formulate linear programming harvest-scheduling problems (objective, variables, constraints) and to solve them using PuLP/HiGHS by default, with optional Gurobi.
- `spatial`: maps aspatial schedules to rasters (e.g., GeoTIFF) for visualization and spatial policy evaluation (hybrid spatial/aspatial workflow).
- 65 • `core/common/financial/forest_helper`: shared utilities for interpolation and curves, convenience helpers for examples, and optional financial indicators.

Data model and workflow. The `ForestModel` class uses discrete time (planning periods) and discrete age classes per dtype. Dtypes are keyed by tuples of theme values (e.g., analysis unit, leading species, site class). For optimization, WS3 aggregates stands into strata defined by dtype and optional zone masks (e.g., management units or policy zones); spatial units (stands, polygons, or raster cells) are only used in the allocation step. Yields are represented by curves/interpolators, and actions (e.g., harvest, planting, fuel treatment, fire) encode operability and transitions (age reset, development type change, growth updates). WS3 builds a state-transition tree per development type across periods. Scheduling can be heuristic (area-control selectors that meet target areas by mask in priority order, e.g., oldest-first) or solver-based via a generic Model I linear program. 75 Indicators are compiled as:

- Action-dependent flows via `compile_product(period, expr, acode=...)` (e.g., harvested volume using a utilization-adjusted expression on total volume yield).
- Inventory stocks via `inventory(period, yname=...)` (e.g., growing stock).

80 *Interoperability.* WS3 imports Woodstock-format text sections (LANDSCAPE, AREAS, YIELDS, ACTIONS, TRANSITIONS) so existing pipelines can be reused without re-encoding model logic. For carbon, `ForestModel` provides `to_cbm_sit` to export Standard Input Table (SIT) configuration and tables consumable by libCBM, with user-provided disturbance-type mappings and per-dtype last-pass-disturbance metadata. Because libCBM ships with parameterizations for multiple countries and ecozones [23], the same export pathways support analyses beyond Canada when users 85 supply the appropriate classifiers and disturbance libraries. Geospatial I/O uses standard formats (shapefile, GeoTIFF). All tabular inputs are plain text (CSV/TSV), and workflows are scripted or notebook-based for reproducibility.

Figure 1 summarizes the WS3 workflow from open inputs through scheduling, carbon analysis, and spatial reporting; Listing 1 outlines the end-to-end case-study workflow used later in the manuscript.

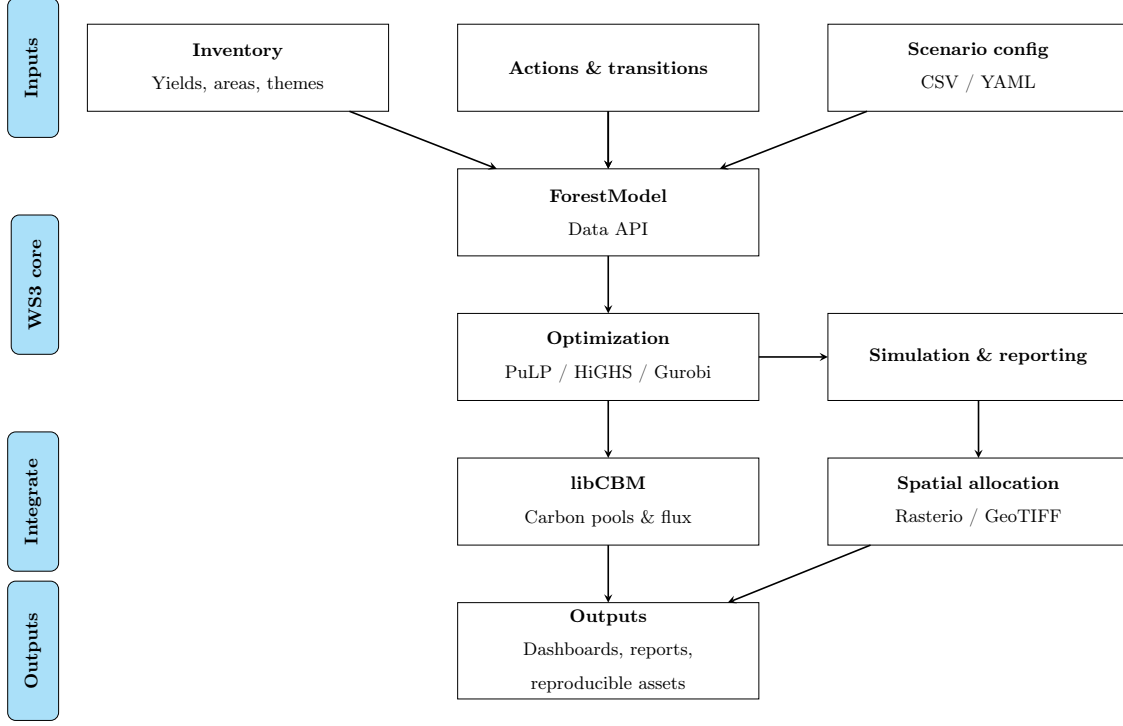


Figure 1: WS3 workflow: categorized inputs feed core scheduling and reporting modules, which integrate with libCBM and spatial allocation utilities before producing reproducible outputs.

The scripted pipeline in Listing 1 underpins the sequential case study documented in Section 3.

We summarize four practical contributions:

- (i) A solver-flexible, path-based Model I generator embedded within a documented forest estate API, with parallel coefficient compilation for large instances.
- (ii) A documented libCBM interface that exports Standard Input Tables and ingests carbon pools and fluxes for integrated accounting.
- (iii) A hybrid aspatial-to-raster allocation workflow that emits reproducible geospatial artefacts from deterministic schedules.

Listing 1: Case-study workflow executed by the reproduction package.

```
def run_case_study(data_root, solver="highs"):  
    inputs = load_inputs(data_root)  
    forest = build_forest_model(inputs)  
    schedule = schedule_harvest(forest, method=solver)  
    sit_tables = compile_to_cbm(schedule)  
    cbm_outputs = run_libcbm(sit_tables)  
    spatial_products = allocate_spatial(schedule)  
    generate_plots_and_tables(schedule, cbm_outputs, spatial_products)  
  
run_case_study("papers/ems")
```

- 100 (iv) A FAIR-aligned reproduction package, including scaling benchmarks on real inventories and archived releases with pinned environments.

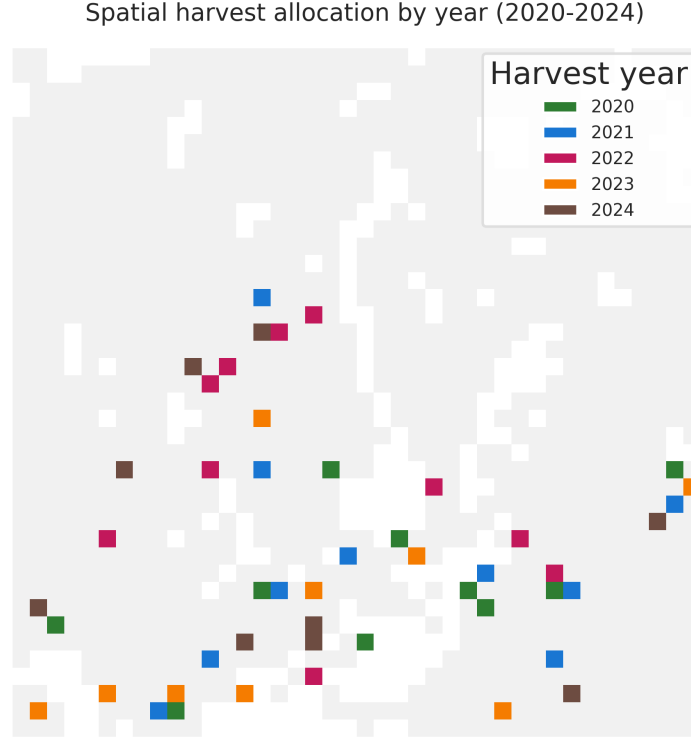


Figure 2: Heuristic schedule allocated to space for the 2020–2024 planning window. Colours denote the first year a cell is harvested, overlaid on the full inventory extent (light grey).

2.2. Optimization formulation (Model I)

WS3 adopts a classic Model I linear programming formulation [21, 15] in which each decision variable selects the proportion of a stratum (a dtype aggregate, optionally intersected with a zone mask) allocated to a feasible root-to-leaf prescription (path) across the planning horizon. Stand identities are aggregated into strata for optimization, and spatial allocation is handled separately.

Let:

I : set of strata (development types; aggregated zones)

J_i : set of feasible prescriptions (paths) for stratum $i \in I$

O : set of outputs (e.g., harvest area, harvest volume, growing stock, habitat)

$O' \subseteq O$: targeted outputs subject to even-flow constraints

T : set of planning periods

$T'_p \subseteq T$: periods on which even-flow for output $p \in O'$ is enforced

Decision variables are proportions:

$$x_{ij} \in [0, 1] \quad \text{for all } i \in I, j \in J_i,$$

meaning “the fraction of stratum i assigned to prescription j ”. If A_i denotes the total area in stratum i , then $A_i x_{ij}$ is the area scheduled on path j . Let μ_{ijot} denote the quantity of output $o \in O$ in period $t \in T$ produced by allocating x_{ij} to path $j \in J_i$ for stratum $i \in I$. Define a reference-period level $y_p := \sum_{i \in I} \sum_{j \in J_i} \mu_{ijpt_p^R} x_{ij}$ for each targeted output $p \in O'$ at $t_p^R \in T$, and ε_p as the allowable even-flow tolerance. With general lower/upper bounds v_{ot}^-, v_{ot}^+ on outputs, the Model I LP is:

$$\text{maximize} \quad \sum_{i \in I} \sum_{j \in J_i} c_{ij} x_{ij} \tag{1}$$

$$\text{subject to} \quad (1 - \varepsilon_p) y_p \leq \sum_{i \in I} \sum_{j \in J_i} \mu_{ijpt} x_{ij} \leq (1 + \varepsilon_p) y_p, \quad \forall p \in O', \forall t \in T'_p \tag{2}$$

$$v_{ot}^- \leq \sum_{i \in I} \sum_{j \in J_i} \mu_{ijot} x_{ij} \leq v_{ot}^+, \quad \forall o \in O, \forall t \in T \tag{3}$$

$$\sum_{j \in J_i} x_{ij} = 1, \quad \forall i \in I \tag{4}$$

$$0 \leq x_{ij} \leq 1, \quad \forall i \in I, \forall j \in J_i. \tag{5}$$

The objective coefficients c_{ij} encode the user-defined value of a path (e.g., total or discounted volume, revenues, multi-criteria scores). Constraints (2) enforce even-flow on targeted outputs, (3) set general lower/upper bounds per period, (4) ensures convex mixing of paths to fully cover each stratum, and (5) bounds variables as proportions. In WS3, μ -coefficients arise from replaying each path through the simulator, calling `compile_product` (action-dependent outputs such as harvest

area and volume) or `inventory` (stocks such as growing stock). The LP matrix is sparse because
120 each path contributes to a limited set of outputs and periods.

Code-to-math mapping. WS3 compiles objective and constraint rows from paths using coefficient functions. Helper routines (e.g., `cmp_c_z`, `cmp_c_caa`, `cmp_c_ci`) traverse path nodes (period, action code, dtype key, age), evaluate μ -like contributions via `compile_product/inventory`, and assemble the LP with `fm.add_problem(...)`. Users select a solver (HiGHS by default; Gurobi
125 optional) and solve via the standard PuLP interface. Upon optimality, WS3 applies the schedule and re-simulates to produce indicators.

Design implications. This path-based Model I approach embeds intertemporal feasibility in each column, separates simulation from optimization, and enables straightforward addition of new outputs/constraints by supplying new coefficient functions. It also aligns with established forest-
130 planning literature while remaining solver-flexible and highly sparse for scalability.

2.3. Implementation

Technology stack. WS3 is implemented in Python and distributed via PyPI. Optimization problems are formulated by WS3 and solved with the open-source HiGHS [18] solver by default, with optional PuLP [28] and Gurobi [17] bindings. Geospatial I/O uses rasterio/fiona/GeoPandas.
135 Documentation (Sphinx) and continuous integration provide API checks, example executability, and unit testing.

libCBM integration. The ForestModel class provides a `to_cbm_sit` method that compiles a CBM Standard Input Table (SIT) configuration and table data structure (i.e., classifiers, inventory, yields, disturbance events, transitions) consumable by libCBM. Users provide a disturbance type
140 mapping and last-pass disturbance metadata to ensure correct dead organic matter (DOM) spin-up. The sequential demonstration runs libCBM for 200 years using the official Python API and documentation [6, 5].

Reproducibility and packaging. The repository includes examples and a reproduction package (papers/ems/repro) with pinned requirements and scripts to generate all figures and tables. Versioned releases are archived on Zenodo [32]. WS3 supports interactive (Jupyter notebooks) and
145 batch (Python scripts) workflows and can be embedded in other software systems as a standard Python package.

2.4. Quality assurance and benchmarking

Automated testing. WS3 ships with a pytest suite that exercises the common, core, financial, forest, and optimization modules (29 tests). The suite runs in approximately 4 s on a Linux workstation (Python 3.12) and is executed for every push and pull request via a GitHub Actions workflow (Ubuntu, Python 3.10) that also regenerates a coverage badge [42]. Style tooling (black, ruff) is bundled in `requirements-dev.txt` and wired into the development makefile, ensuring consistent formatting before release. Beyond unit tests, WS3 has been exercised in multiple operational planning projects over the past decade, providing repeated end-to-end validation of core workflows.

Input stewardship and validity. WS3, like its Woodstock heritage, makes no assumptions about inventories, yields, operability, objectives, or constraints. The space of “valid input” is project- and policy-specific and inherently high dimensional; exhaustive auto-validation is out of scope. WS3 provides targeted checks in high-leverage paths (e.g., fallback to template transitions when age-specific rules are missing; basic range/shape sanity checks) and clear documentation of required structures for inventories, yields, actions, and transitions. Responsibility for input quality and modeling assumptions rests with qualified analysts; WS3 is a professional framework rather than a prescriptive application.

Determinism and reproducibility. Given identical inputs, WS3 produces identical outputs. Minor nondeterminism is limited to optional solver seeds and randomized block ordering in the spatial allocation utility; both are controllable and fixed in the reproduction scripts. Consequently, statistical variance on deterministic time series is not meaningful; we report solved status, model scale, and runtime/throughput metrics instead. The reproduction package pins versions and seeds to yield byte-identical artifacts across runs, reflecting best practices for computational reproducibility [36].

Verification against Woodstock. WS3 was originally verified against Woodstock (2015) by replaying identical datasets and comparing action-specific flows and stocks across periods, yielding functionally equivalent outputs for the implemented subset of features. Known departures from Woodstock semantics are flagged in code and documentation. Ongoing development follows a trust-but-verify pattern: changes to core loops are regression-tested against known-good benchmarks before release. The Woodstock parity check included in this manuscript is therefore a transparent, reproducible illustration rather than an exhaustive proof of equivalence.

Case-study repro checks. The EMS reproduction package encapsulates the sequential WS3→libCBM

pipeline in `make_repro.sh`. Inside the project’s reference LXD (Linux Containers) environment (Ubuntu 24.04 on dual Xeon 6254 CPUs, 72 logical cores, 768 GB PC-23400 RAM) the script completes in 6.1 s, re-creating Figures 6–7, Figure 2, and the tables stored in `papers/ems/tables`. The workflow is deterministic: running `generate_case_study.py` a second time produces byte-identical CSV and PNG assets, providing an integration test for the scheduling, SIT export, and libCBM coupling.

Performance spot checks. The heuristic scheduler used in the case study produces a 100-year plan in under a second, and the subsequent 200-year libCBM simulation dominates total runtime (~5 s). Larger linear programs formulated through `ws3.opt` benefit from HiGHS (default) or optional Gurobi: internal regression notebooks (e.g., Examples 030/040) routinely solve tens-of-thousands-of-stand path-based instances within minutes using HiGHS, while the optional Gurobi backend shortens solve time further via its advanced presolve and feasibility tools. Raster allocations scale linearly with the number of cells processed per period and are trivially parallelizable.

Optional scalability experiment. To characterize scaling on real inventories, the reproduction package optionally integrates a DataLad-hosted benchmark dataset comprising five non-overlapping timber supply areas (TSAs) in northeastern British Columbia (the same study region as Boisvenue et al. 2). Enabling the optional step (`RUN_SCALING=1`) fetches the dataset and executes `papers/ems/repro/run_scaling_benchmarks.py`, which sorts TSAs by complexity and evaluates them individually and cumulatively (`tsa08`, `tsa08+tsa40`, ..., `tsa08+tsa40+tsa41+tsa24+tsa16`). The scripted benchmark always runs the heuristic scheduler on a single worker—reflecting the implementation’s serial design—and records sequential spatial-allocation runtimes using `ForestRaster`. On the reference container the heuristic schedules complete in 1.3–31.4 s while spatial allocation requires 46–656 s as problem size grows from 4.7 k to 37.7 k dtype–age pairs. Setting `RUN_LP=1` extends the run to model-building benchmarks for the deterministic Model I LP: WS3 constructs the LP with 1 and 16 workers (parallel coefficient compilation) and solves it with matching HiGHS thread counts, logging build/solve time, solver status, and stage-wise memory footprints. LP builds scale from 5.1 s (4.7 k dtype–age pairs) to 92.8 s (37.7 k dtype–age pairs) with single-worker compilation, while the 16-worker mode trims build time to 59 s at the cost of higher peak RSS (2.5 GB versus 1.8 GB). Because large path-based LPs routinely exceed tens of gigabytes of RAM on even larger inventories, the LP measurements are opt-in and intended for suitably provisioned systems. All metrics are written to `perf_scaling.csv` with deterministic seeds so that regenerated figures

210 and tables remain byte-identical; Figures 3–5 and Table 1 summarise the resulting runtimes and memory profiles (Figures 3 and 4).

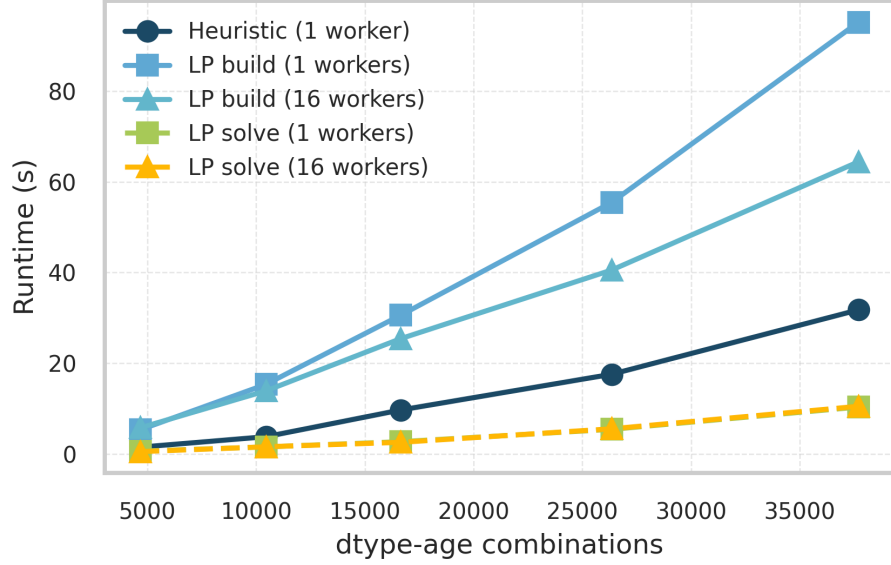


Figure 3: Scheduling runtime versus problem size, measured by the number of development type (dtype) by age combinations ($n_{\text{dtype-age}}$). Heuristic runs are single worker; LP build/solve lines appear when `RUN_LP=1`.

3. Illustrative use-case demonstrations

We present a two-stage sequential pipeline that schedules harvest in WS3 and then evaluates carbon dynamics using libCBM. The workflow replicates example 031_ws3_libcbm_sequential-
 215 builtin.ipynb to ensure reproducibility.

Study design. We use the public example dataset `tsa24_clipped`. It is intentionally modest in size so that the full pipeline (scheduling, carbon accounting, and spatial allocation) can be executed and audited on standard hardware, and it mirrors the tutorial dataset distributed with WS3 to ensure transparent, reproducible inputs. Inputs are stored as Woodstock-format text files
 220 (landscape, areas, yields, actions, transitions) under `examples/data/woodstock_model_files_tsa24_clipped` and imported into a `ForestModel`. We schedule harvest using a self-parameterizing area-control heuristic to produce an aspatial action schedule over a 100-year horizon (10 periods of

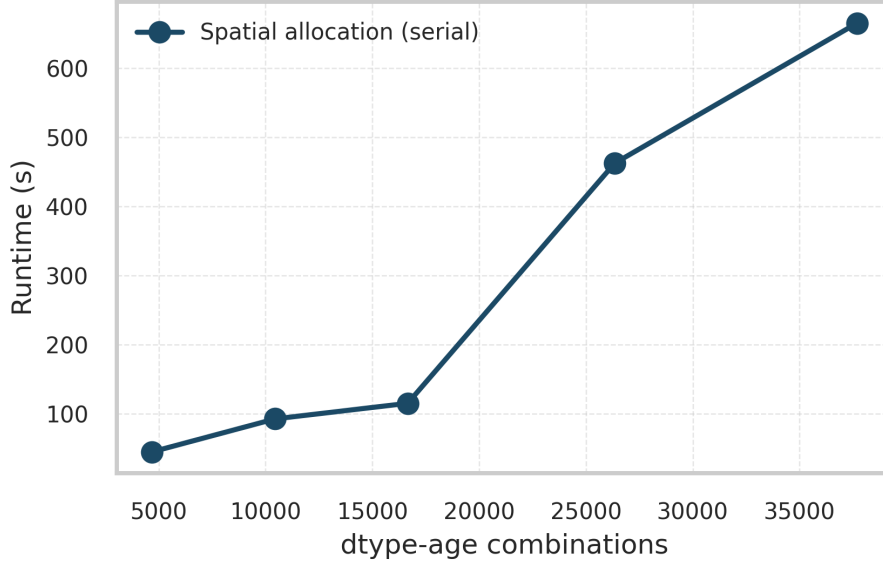


Figure 4: Spatial allocation runtime (serial **ForestRaster**) versus problem size after applying the heuristic schedule; problem size counts dtype–age combinations ($n_{\text{dtype-age}}$).

10 years), then export a libCBM Standard Input Table (SIT) to simulate annual carbon stocks and fluxes.

225 *Data and study area.* The `tsa24_clipped` dataset is a clipped subset of a Timber Supply Area inventory used for demonstration. The initial inventory enumerates development types (dtypes) by theme tuples (e.g., analysis unit, species, site), with yield curves for total volume (`totvol`) and species-volume splits (`swdvol`, `hwdvol`). Actions and transitions define operability and state changes (e.g., harvest resets age).

230 *Model setup and scheduling.* We initialize the model with `base_year=2020`, `horizon=10`, `period_length=10y`, and `max_age=1000`. After importing sections and initializing areas, we add a null action and reset actions. The area-control heuristic proceeds period by period: for each target mask (default: AU-wise THLB), it computes a target area using area-weighted mean CMAI age and then applies the harvest action to the oldest operable ages within the masked dtype set
 235 until the target is met (or inventory is exhausted). The resulting schedule is `aspatial`, recording treatment area by dtype, age, and period. Utilization is set to 0.85 in the volume expression used

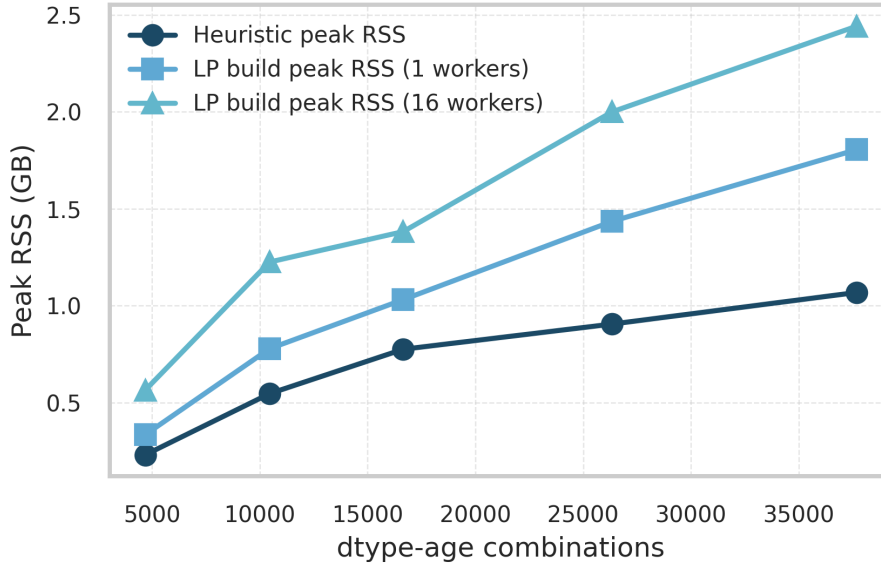


Figure 5: Peak resident memory for heuristic scheduling (serial) and LP model builds (1 and 16 workers) as problem size increases in terms of dtype-age combinations.

for compiled flows. The schedule is compiled and applied to produce period-by-period flows and stocks.

libCBM coupling and metrics. We define a disturbance-type mapping for actions (harvest→Clearcut harvesting without salvage; fire→Wildfire) and assign `last_pass_disturbance` per dtype to ensure consistent DOM spin-up. Using `ForestModel.to_cbm_sit` with softwood and hardwood volume names (`swdvol`, `hwdvol`), `admin_boundary = "British Columbia"`, and `eco_boundary = "Montane Cordillera"`, we export SIT config and tables. We then run libCBM for `n_steps=200` years, aggregating annual carbon stocks (biomass, DOM, total ecosystem).

Outputs. We report time series of harvested area/volume and growing stock from WS3, and annual carbon stocks from libCBM. Spatial allocation then maps the aspatial schedule onto raster cells using the `ForestRaster` utility to generate a disturbance footprint while preserving the aspatial totals. The reproduction package generates the figures and tables used below.

The full, deterministic workflow is scripted in `papers/ems/repro/generate_case_study.py`, which produces the flows table and carbon stocks table:

Table 1: Peak resident memory during heuristic scheduling and LP model builds on the reference container. $n_{\text{dtype-age}}$ counts development type (dtype) by age combinations; LP measurements report the maximum RSS observed while compiling the Model I matrix with 1 or 16 workers, while spatial allocation runs serially.

Combo	$n_{\text{dtype-age}}$	Heuristic peak RSS (GB)	LP peak RSS (GB; 1 worker)	LP peak RSS (GB; 16 workers)
tsa08	4 682	0.23	0.33	0.56
tsa08+tsa40	10 453	0.52	0.78	1.22
tsa08+tsa40+tsa41	16 653	0.77	1.03	1.33
tsa08+tsa40+tsa41+tsa24	26 346	0.87	1.38	1.87
tsa08+tsa40+tsa41+tsa24+tsa16	37 691	1.07	1.81	2.51

- `papers/ems/tables/scenario_flows.csv`: harvest area, harvest volume, and growing stock by period.
- `papers/ems/tables/annual_carbon_stocks.csv`: annual biomass, DOM, and total ecosystem carbon.

255 Figures 6 and 7 visualize the resulting flows and carbon stocks. The entire pipeline, including the workflow overview (Figure 1), workflow pseudocode (Listing 1), and the spatial allocation example (Figure 2), is reproducible via `papers/ems/repro/make_repro.sh`. The libCBM run length (200 years) matches the example and can be adjusted.

Unless stated otherwise, areas are reported in hectares (ha) and volumes in cubic metres (m^3).

260 *Results.* The heuristic scheduler delivers a 10-period harvest plan whose flows stay within $\sim 10\%$ of their period means without any explicit even-flow constraint: harvest area averages 106.8 ha and ranges from 103 to 114 ha, while harvest volume averages 15.3 thousand m^3 with a 14.1–18.5 thousand m^3 span (Table 2; Figure 6). Growing stock declines smoothly from 135 thousand to 82 thousand m^3 (consistent with the 0.85 utilization factor), documenting the state trajectory exported to libCBM. In the carbon stage, libCBM aggregates show biomass pools increasing from 265 2.7×10^5 tC to 3.1×10^5 tC over 200 years, while DOM remains within $1.7\text{--}1.9 \times 10^5$ tC, yielding a total ecosystem carbon gain of 42 ktC (Figure 7). These values are illustrative; the key contribution is the deterministic, auditable pipeline that connects scheduling outputs to carbon accounting and spatial allocation with byte-identical artefacts. The summary tables `scenario_flows.csv` and

Table 2: Sequential demonstration parameters and configuration (WS3 → libCBM pipeline).

Setting	Value
Dataset	<code>tsa24_clipped</code> (Woodstock-format inputs; <code>examples/data/woodstock_model_files</code>)
Base year	2020
Planning horizon	10 periods (100 years)
Period length	10 years
Max age class	1000 years
Yield names	<code>totvol</code> (total volume), <code>swdvol</code> (softwood), <code>hwdvol</code> (hardwood)
Scheduling method	Area-control heuristic (priority-queue, oldest-first); utilization 0.85
Disturbance mapping	harvest→Clearcut harvesting without salvage; fire→Wildfire (CBM)
CBM export	<code>ForestModel.to_cbm_sit</code> (admin: British Columbia; eco: Montane Cordillera)
CBM run length	200 annual steps
Outputs	<code>scenario_flows.csv</code> (period, harvest_area_ha, harvest_volume_m3, growing_stock_m3); <code>annual_carbon_stocks.csv</code> (annual pools)

270 `annual_carbon_stocks.csv` (`papers/ems/tables`) retain the raw outputs for external verification.

Reproducibility. This demonstration is intended to be transparent and portable; all input data reside in `examples/data/woodstock_model_files_tsa24_clipped`. The spatial allocation step renders the raster-based harvest allocation shown in Figure 2, derived directly from the aspatial schedule via the `ForestRaster` utility. The folder `papers/ems/repro` contains `make_repro.sh`, which
275 provisions a clean virtual environment, installs pinned dependencies (including the optional libCBM extra via `pip install ws3[cbm]`), and runs deterministic scripts that regenerate Figures 1–7 and the accompanying CSV tables. Outputs are byte-identical on the reference container and stable under the pinned dependencies; the exact environment is preserved in the Zenodo-archived release.

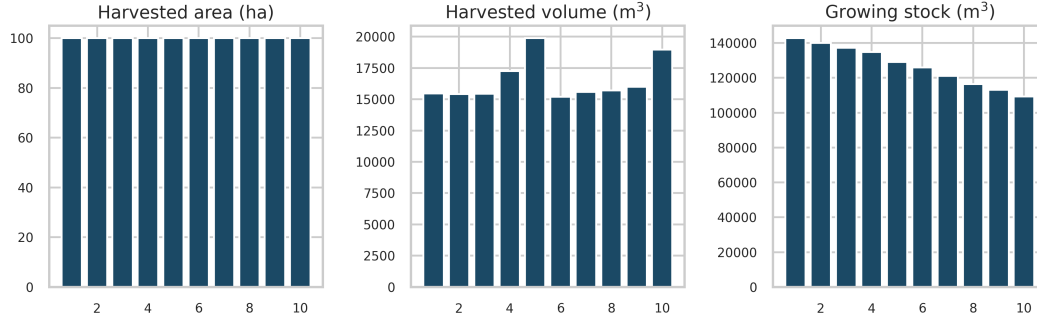


Figure 6: WS3 results: harvested area, harvested volume (utilization-adjusted), and growing stock over planning periods.

Broader example library and carbon-aware optimization

Beyond the sequential carbon-accounting demonstration, WS3 includes executable notebooks that act as tutorials and validation exercises, from synthetic cases to policy-oriented analyses. Example 040 implements the Neilson hack and compares aggregated WS3 carbon indicators with libCBM in a no-disturbance scenario. Example 041 illustrates carbon-aware optimization by embedding carbon prices/constraints in the LP and evaluating schedules with `libcbm_py`.

4. Impact and reusability

WS3 has been used in research on bioenergy [7], value-creation potential and supply modelling [34, 33], and climate impact assessment [40, 22, 46]. It integrates with the SpaDES ecosystem for spatial simulation [8] and has been deployed as a backend in decision-support applications. The MIT license, modular API, and PyPI distribution support reuse across jurisdictions.

Major use-cases (2015–2025).

- Strategic wood supply and bioenergy planning in mixed-wood forests: optimization and scenario analysis of value-creation options [7].
- Hybrid simulation–optimization for value indicators and model retrofitting: methods and transfer to production-style analyses [34, 33].

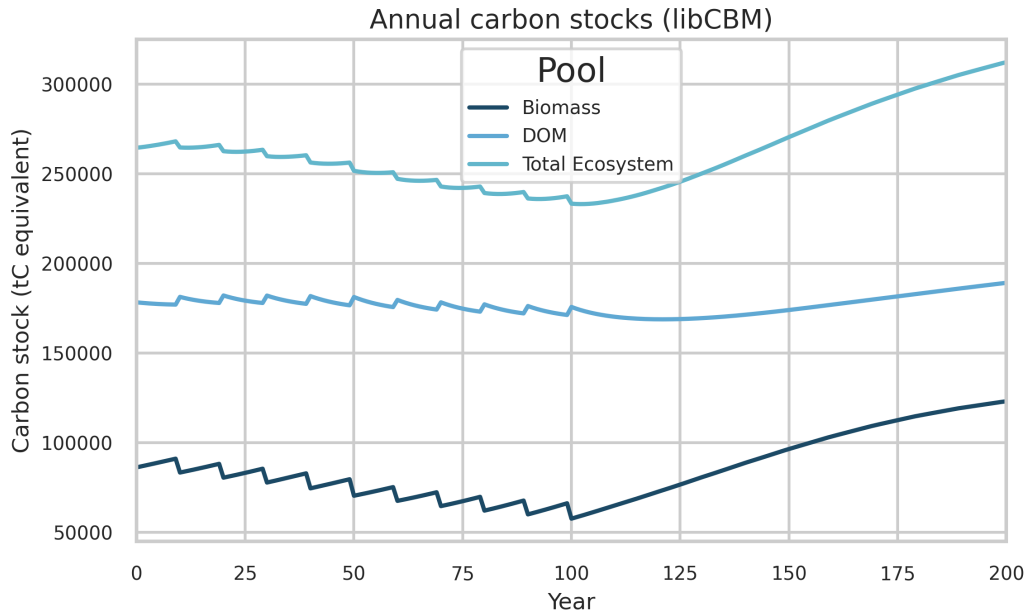


Figure 7: libCBM results: annual carbon stocks (biomass, dead organic matter, total ecosystem) over 200 years.

- Climate mitigation and carbon-aware planning in British Columbia: sequential WS3→libCBM pipelines and optimization under policy constraints; graduate theses and applied demonstrations [22, 46, 45].
- Decision-support prototypes for nature-based solutions: avoided fire and avoided harvest workflows (Examples 050/060) with net-emission indicators derived from libCBM.
- Integration with spatial simulation: `spades_ws3` module bridging WS3 with SpADES for disturbance and landscape dynamics studies [8, 44].
- Application backends: WS3 used in web-based decision-support systems (e.g., `ecotrust-dss`) for scenario configuration and reporting [43].
- Ongoing initiatives: CCCANDiES (integrated carbon and ecosystem services; in progress) and mining-sector nature-based solutions [13].

Interoperability is supported through tabular/text inputs, raster outputs, and the libCBM linkage, which allow WS3 to slot into existing analytical pipelines. The reproduction package, unit

tests, and continuous integration (CI) pipelines facilitate method transfer and support open scientific practice [19, 26]. The architecture is general and has been applied to studies of sustainable yield, carbon accounting, biodiversity, and policy trade-offs [2].

Reusability features.

- Openness and packaging: MIT license, PyPI distribution, versioned releases archived on Zenodo [32]; complete examples and tutorials.
- Interoperability: Woodstock-format import for legacy/industry pipelines; libCBM SIT export for carbon; standard geospatial formats (GeoTIFF, shapefile).
- Reproducibility: deterministic reproduction package (`papers/ems/repro`) that regenerates all figures/tables; continuous integration (CI) runs tests and validates examples.
- Extensibility: modular API for adding outputs/constraints and new scheduling logic (heuristic or LP-based); solver-flexible (HiGHS by default; Gurobi optional).
- Integration pathways: SpaDES ecosystem via `spades_ws3`; backend for decision-support applications and web services.
- Education and onboarding: progressive example notebooks (from 010 to 060) facilitate training and method transfer.

Example 040 demonstrates a stand-level optimization workflow with explicit integration of carbon dynamics using `libcbm_py`. The workflow shows how WS3’s optimization layer and libCBM coupling can support carbon-aware planning experiments while leaving the precise objective formulation to the user.

5. Comparison to alternatives

Parity with Woodstock. To reassure readers that WS3 implements standard inventory accounting correctly, we report a parity check against a prescriptive Woodstock run using the same toy dataset and an identical action schedule. This test is intended as a transparent, reproducible illustration rather than an exhaustive proof of equivalence. The accompanying CSV artifact `papers/ems/tables/woodstock_parity.csv` provides total harvest area, harvest volume, and growing stock,

Table 3: Comparison of WS3 with selected tools.

Tool	Scope	License	Optimization	Spatial	Carbon	Region
WS3	Landscape	Open (MIT)	Built-in (PuLP; op- tional Gurobi)	Hybrid aspatial-to- raster	Built-in libCBM link- age	General
Woodstock	Landscape	Proprietary	Built-in	Extensions; connectors	External inte- grations	General
Patchworks	Landscape	Proprietary	Built-in	Spatial plan- ning emphasis	External inte- grations	General
SpaDES	Simulation framework	Open	External pack- ages	Fully spatial (agent-based, raster)	Via modules	General
SIMFOR	Stand-level	Open	Limited/none	N/A	Limited/none	General
Heureka	Landscape	Restricted (mixed)	Built-in	Spatial capa- bilities	External inte- grations	EU-focused
JLP (MELA)	Landscape	Restricted (legacy)	Built-in	Spatial exten- sions	External inte- grations	Finland- focused

along with WS3-relative percent differences (computed as $100 \cdot (\text{WS3} - \text{Woodstock}) / \text{Woodstock}$). For transparency, period-wise parity summaries for harvest area and volume can be regenerated via the reproduction scripts and are included in the reproduction package. (Figure 9) Exact or near-exact agreement is expected given identical inputs.

Table 4: WS3 vs. Woodstock parity on the toy dataset (identical schedule). Values are loaded from the accompanying CSV artifact.

Metric	WS3	Woodstock	WS3 vs Woodstock (%)
Total harvest area (ha)	1,000.00	1,000.00	0.00
Total harvest volume (m ³)	164,838.08	164,838.08	0.00
Total growing stock (m ³)	1,269,692.15	1,269,692.15	0.00

WS3 differs in that it provides an open Python API, a documented libCBM integration, and a hybrid spatial/aspatial workflow intended for reproducible analyses. It is a framework rather than a GUI decision-support suite; spatial adjacency/opening-size constraints and road-network planning are outside its scope. SIMFOR operates at stand level, whereas WS3, Woodstock, Patchworks, Heureka, and JLP target landscape-scale planning with varying degrees of spatial explicitness and

availability [38, 41, 8, 20, 29, 39].

Comparison narrative. Proprietary landscape systems such as Woodstock and Patchworks provide
345 mature optimization and spatial planning capabilities, including native support for adjacency and
block constraints, but their licensing and closed codebases limit transparent methods development,
automated testing, and reproducibility at scale. SpaDES is an open simulation framework oriented
to spatial processes and agent-based dynamics; it does not natively provide a solver-backed harvest
scheduling layer but can interoperate with WS3 via `spades_ws3` for hybrid workflows. Heureka and
350 JLP are landscape-oriented and include optimization, yet are region-specific or legacy-maintained,
which constrains adoption beyond their primary jurisdictions.

Carbon-aware planning and optimization has an active literature that couples timber supply
models with carbon accounting and evaluates mitigation trade-offs under alternative schedules and
policies. Early integrations such as CO2Fix paired with timber supply modelling illustrate carbon-
355 informed planning workflows [30]. More recent sector and landscape studies quantify mitigation
potential and carbon capacities under harvest scenarios [40, 2], and graduate theses in British
Columbia explicitly evaluate climate objectives in forest planning optimization contexts [22, 46].
WS3 provides a reproducible, open framework for constructing these workflows while keeping the
scheduling engine transparent and auditable.

360 WS3 integrates: (i) a solver-flexible, path-based Model I LP generator embedded in an open
forest estate API; (ii) a documented linkage to libCBM for carbon stocks and fluxes; and (iii) a
hybrid spatial/aspatial pipeline with standard geospatial I/O. This design supports reproducible
workflows in notebooks and scripts and allows integration into larger research and decision-support
ecosystems. Where spatially explicit optimization is required, WS3 can provide schedules and
365 carbon accounting inputs to specialized spatial tools; carbon-related objectives or constraints can be
added when users supply coefficients, but they are not required for the core workflow demonstrated
here.

6. Discussion

WS3 is intended as an open, reproducible alternative to proprietary forest-estate planning sys-
370 tems. Its contribution is primarily informatics: a transparent, scriptable workflow that integrates
scheduling, carbon accounting, and spatial reporting, rather than a new optimization formulation.

6.1. Scope and limitations

WS3 focuses on deterministic, aspatial forest estate formulations. The Model I LP implementation is well suited to even-flow policies, policy bounds, and value maximization problems, yet it inherits the classic limitations of the formulation: stochastic or endogenous disturbances cannot be represented faithfully without departing from linearity, and non-linear objectives or constraints require either linearisation or external wrappers. In practice, we rely on the heuristic scheduler to interleave random disturbances or sensitivity analyses when scenario logic demands it, and reserve the LP for static policy experiments.

Spatial representation is delivered through a post-hoc raster allocation step that respects operability and transitions but does not enforce spatial adjacency, road-building, or block-shape constraints. Those dynamics fall outside the scope of WS3. Users requiring spatial optimisation should therefore treat WS3 as a scheduling and accounting engine that feeds specialised spatial tools.

In this manuscript, carbon accounting is applied post hoc to schedules produced by the LP or heuristic workflows. The framework can be extended to incorporate carbon-related objectives or constraints when users supply the necessary coefficients, but those formulations are not the focus of the present study.

Finally, large LP instances involve substantial memory pressure—tens of gigabytes for the largest benchmarks—because the column generation compiles full path matrices. The reproduction package flags the LP scaling experiment as opt-in so that teams can execute it on appropriately provisioned infrastructure. All supporting assets, however, remain deterministic and auditable even when the LP portion is skipped.

7. Conclusions and future work

WS3 provides an open, reproducible framework for integrating harvest scheduling, simulation, carbon accounting, and spatial reporting. By coupling solver-backed scheduling with libCBM via a documented interface, WS3 supports transparent Model I analyses that can be inspected, reproduced, and extended.

Limitations remain: results depend on the quality of user-supplied yields, transition rules, and carbon parameters; stochastic disturbances and climate-forced growth uncertainty require external scenario ensembles; and large LP instances can require substantial memory. The reproduction package flags these constraints and provides deterministic assets for audit and reuse.

Future work will prioritize uncertainty handling and spatial extensions, guided by community contributions to the open codebase.

Reproducibility. All figures and tables are generated by scripts in `papers/ems/repro`. The `make_repro.sh` script creates an isolated environment, installs pinned dependencies (`requirements.txt`), and runs `generate_diagrams.py`, the spatial allocation workflow (`generate_spatial_allocation.py`), and the sequential demonstration script (`generate_case_study.py`). Outputs are written to `papers/ems/figs` and `papers/ems/tables`. The exact versions are preserved via the Zenodo-archived release.

FAIR compliance. Table 5 summarises the EMS FAIR/software checklist submitted with the manuscript (see supplementary CSV `papers/ems/tables/fair_checklist.csv`). WS3 remains Findable through its Zenodo DOI archive and indexed repositories, Accessible under the MIT license with public code/data artefacts, Interoperable via open tabular and geospatial formats plus Woodstock/libCBM linkages, and Reusable through pinned environments, continuous-integration (CI)-tested workflows, and full reproduction scripts [32].

Table 5: EMS FAIR/software checklist summary for WS3.

Principle	Checklist focus	Evidence / artefact
Findable	Persistent identifier; indexed repository	Zenodo DOI 10.5281/zenodo.17331213 ; tagged GitHub releases; PyPI project metadata
Accessible	License and public access	MIT license (CITATION.cff, LICENSE); public GitHub repository; PyPI wheels; reproduction package assets bundled in repository
Interoperable	Open formats and connectors	Woodstock LAN/ARE/YLD importers; libCBM SIT export; tabular CSV, GeoTIFF, and shapefile outputs; documented workflow in Figure 1
Reusable	Documentation, provenance, validation	ReadTheDocs site; notebooks/examples; continuous-integration (CI)-tested <code>pytest</code> suite; deterministic scripts in <code>papers/ems/repro</code> ; submission checklist metadata

CRediT authorship contribution statement

G.E.P.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review & editing, Visualization, Supervision, Project administration, Funding acquisition.

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Data availability

Data availability: Source code, configuration files, and reproduction assets are archived at <https://doi.org/10.5281/zenodo.17331213>. All scripts used to generate the figures and tables are included; reuse is permitted under the MIT License.

Declaration of competing interest

Competing interests: The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used the OpenAI GPT-5 and GPT-5-Codex models to assist with language polishing and format compliance. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Example 040: CBM vs WS3 carbon indicators

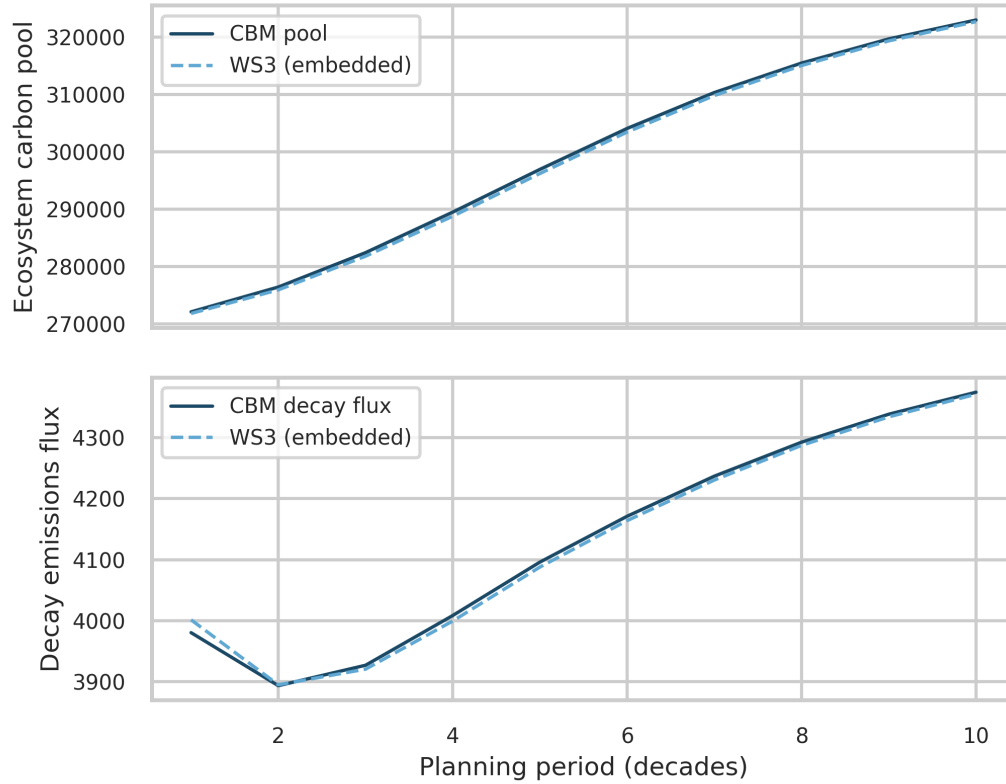


Figure 8: Neilson-hack demonstration (Example 040): comparison of libCBM (solid) and WS3-embedded (dashed) carbon indicators after bootstrapping pools/fluxes into WS3 as yield curves. Top: ecosystem carbon pools; Bottom: aggregate decay emissions flux. In a no-disturbance scenario, the aggregated indicators match closely; with harvesting, flux gaps reflect known limitations of the Neilson hack due to CBM DOM spin-up assumptions mapping time to age (Figure 8).

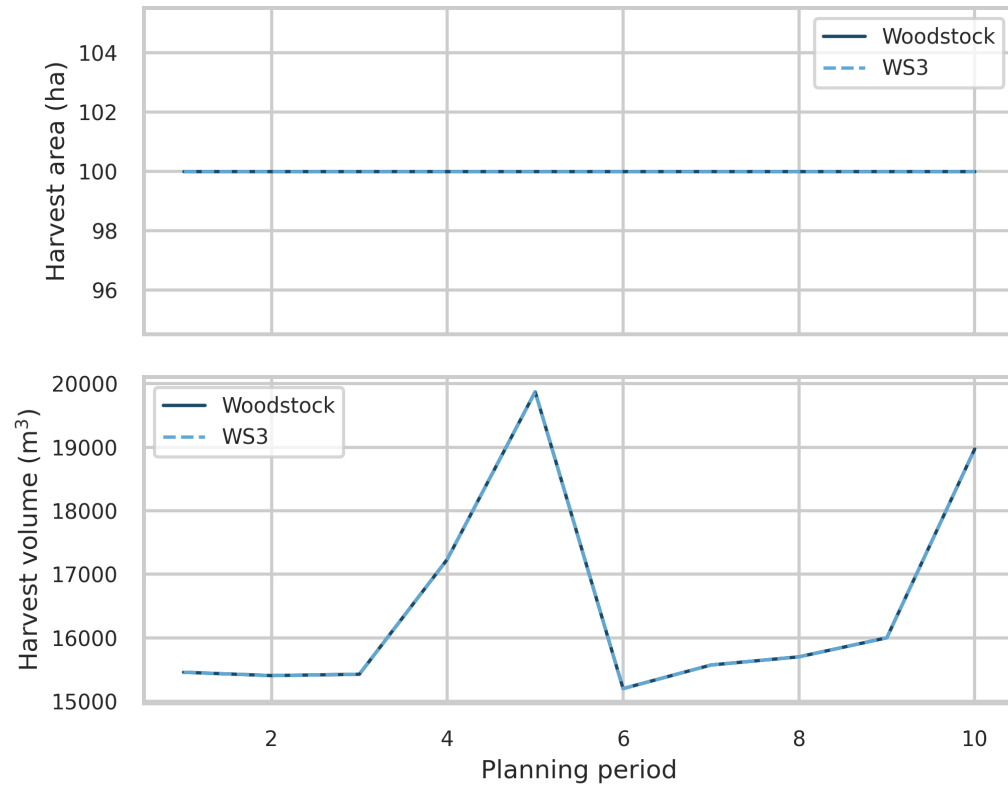


Figure 9: Period-wise parity of harvested area and harvested volume between WS3 and Woodstock on the toy dataset (identical schedule). Values are loaded from the accompanying CSV artifact.